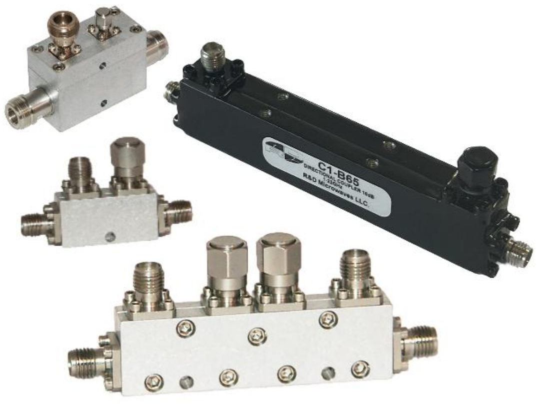
第三章 耦合器理论及设计



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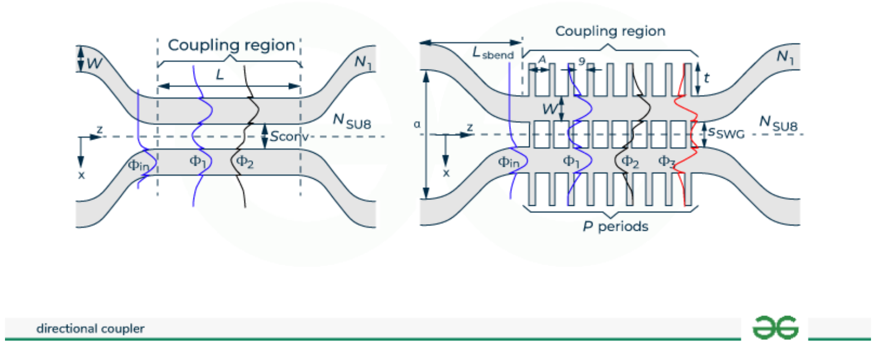
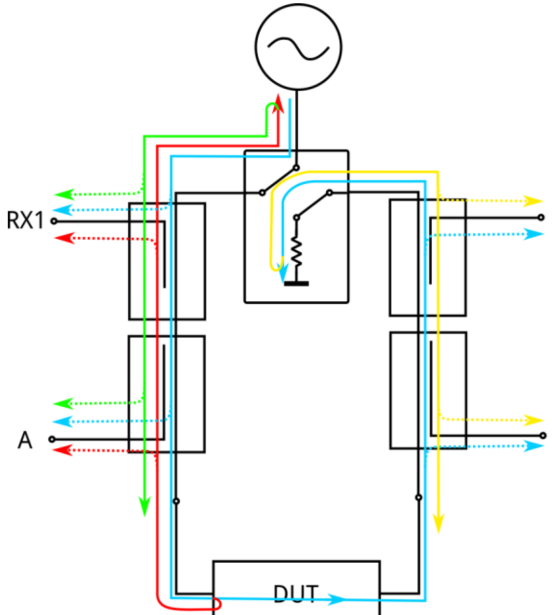
2026-05

耦合器

- 微波耦合器（Microwave Coupler/Directional Coupler）是一种重要的射频/微波无源四端口器件，主要用于将微波信号主干通道的功率按一定比例（通常是不等功率）提取出一小部分用于信号监测、功率分配、隔离或反射测量。其主要特征是具有方向性，即将主线上的信号仅耦合到副线的一个输出端。





网络分析仪原理图

耦合器

$$[S]=\begin{bmatrix} 0 & S_{12} & S_{13} & S_{14} \\ S_{12} & 0 & S_{23} & S_{24} \\ S_{13} & S_{23} & 0 & S_{34} \\ S_{14} & S_{24} & S_{34} & 0 \end{bmatrix}.$$

互易性： $[S]=[S]^t$

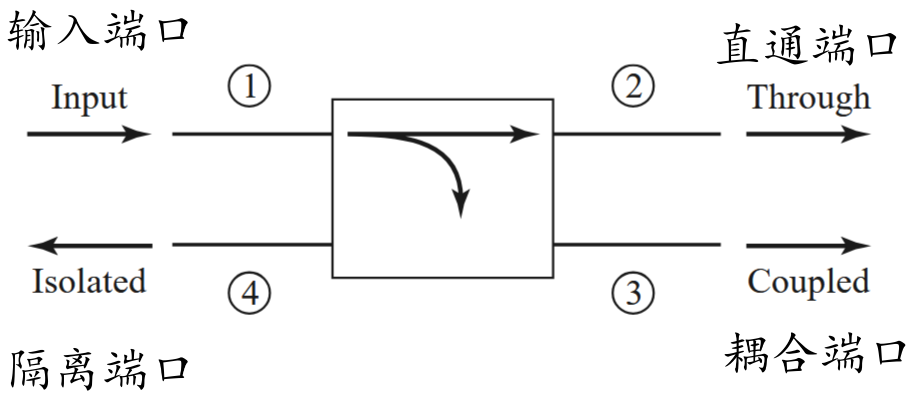
无耗性： $[S]^t[S]^*=[U]$ ⟹ $\sum_{k=1}^{N} S_{ki}S_{kj}^*=\delta_{ij}$, for all i, j,

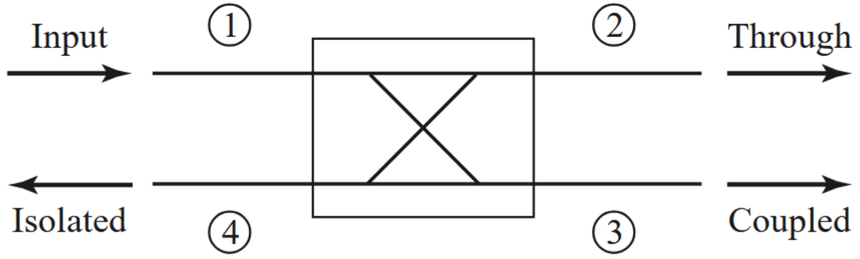
$$[S]=\begin{bmatrix} 0 & \alpha & j\beta & 0 \\ \alpha & 0 & 0 & j\beta \\ j\beta & 0 & 0 & \alpha \\ 0 & j\beta & \alpha & 0 \end{bmatrix}$$

对称网络

$$[S]=\begin{bmatrix} 0 & \alpha & \beta & 0 \\ \alpha & 0 & 0 & -\beta \\ \beta & 0 & 0 & \alpha \\ 0 & -\beta & \alpha & 0 \end{bmatrix}$$

反对称网络

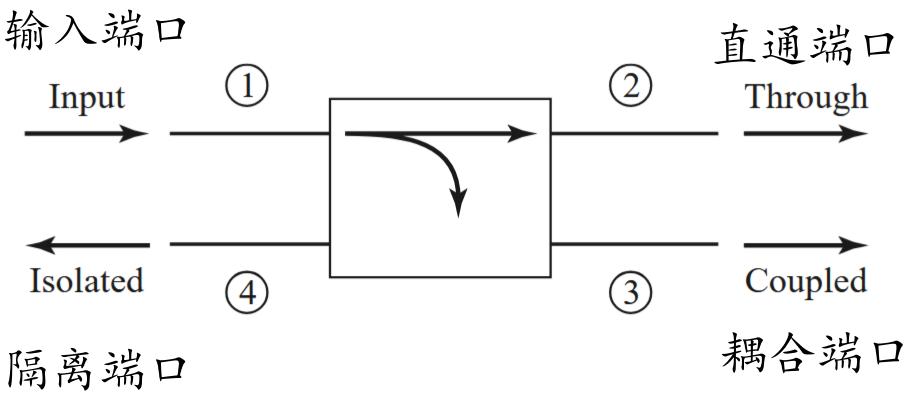


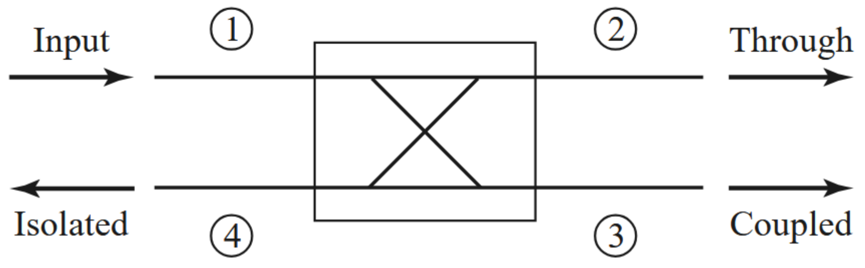


耦合器

$$[S]=\begin{bmatrix}0 & \alpha & j\beta & 0\\ \alpha & 0 & 0 & j\beta\\ j\beta & 0 & 0 & \alpha\\ 0 & j\beta & \alpha & 0\end{bmatrix}$$

$$[S]=\begin{bmatrix}0 & \alpha & \beta & 0\\ \alpha & 0 & 0 & -\beta\\ \beta & 0 & 0 & \alpha\\ 0 & -\beta & \alpha & 0\end{bmatrix}$$





耦合度：

$$\text{Coupling} = C = 10\log\frac{P_1}{P_3} = -20\log\beta \text{ dB},$$

方向性：

$$\text{Directivity} = D = 10\log\frac{P_3}{P_4} = 20\log\frac{\beta}{|S_{14}|} \text{ dB},$$

隔离度：

$$\text{Isolation} = I = 10\log\frac{P_1}{P_4} = -20\log|S_{14}| \text{ dB},$$

插入损耗：

$$\text{Insertion loss} = L = 10\log\frac{P_1}{P_2} = -20\log|S_{12}| \text{ dB}.$$

耦合度C：有多少功率通过耦合进入耦合端口

方向性D：衡量耦合器隔离前向和后向波的能力

隔离度I：衡量耦合器的隔离端口的能力

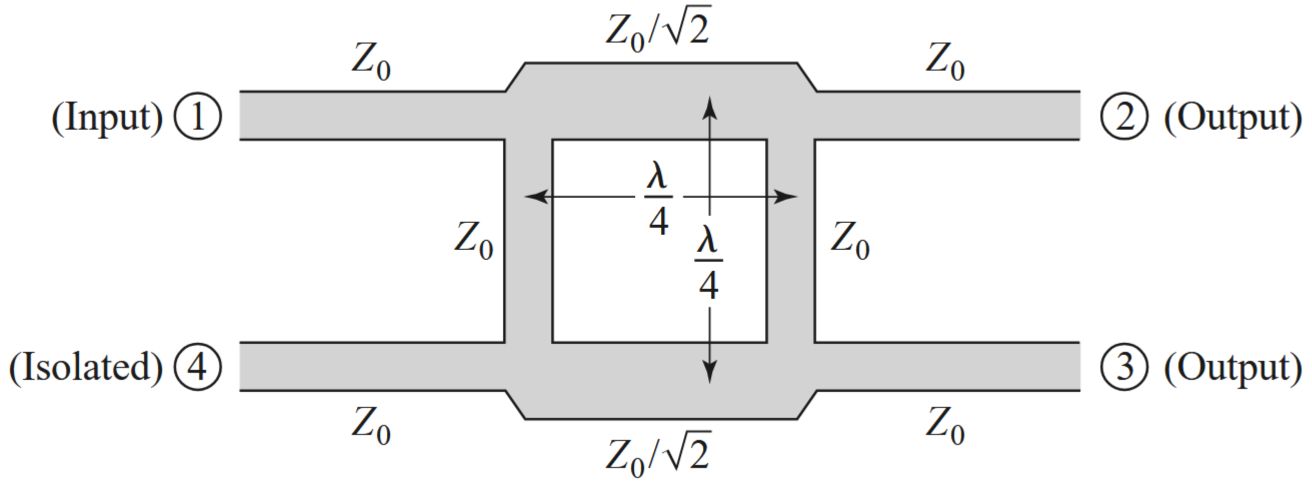
插入损耗L：直通端口的功率损耗了多少

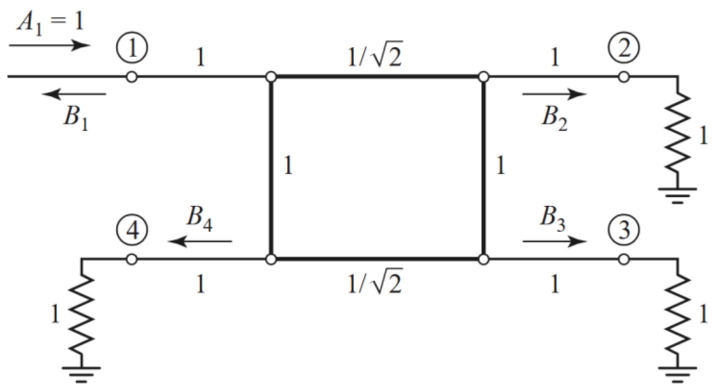
CASE STYLE: HT2679

ZCDC10-01263-S+.pdf

支线耦合器（正交混合耦合器）

$$[S] = \frac{-1}{\sqrt{2}}\begin{bmatrix} 0 & j & 1 & 0 \\ j & 0 & 0 & 1 \\ 1 & 0 & 0 & j \\ 0 & 1 & j & 0 \end{bmatrix}$$

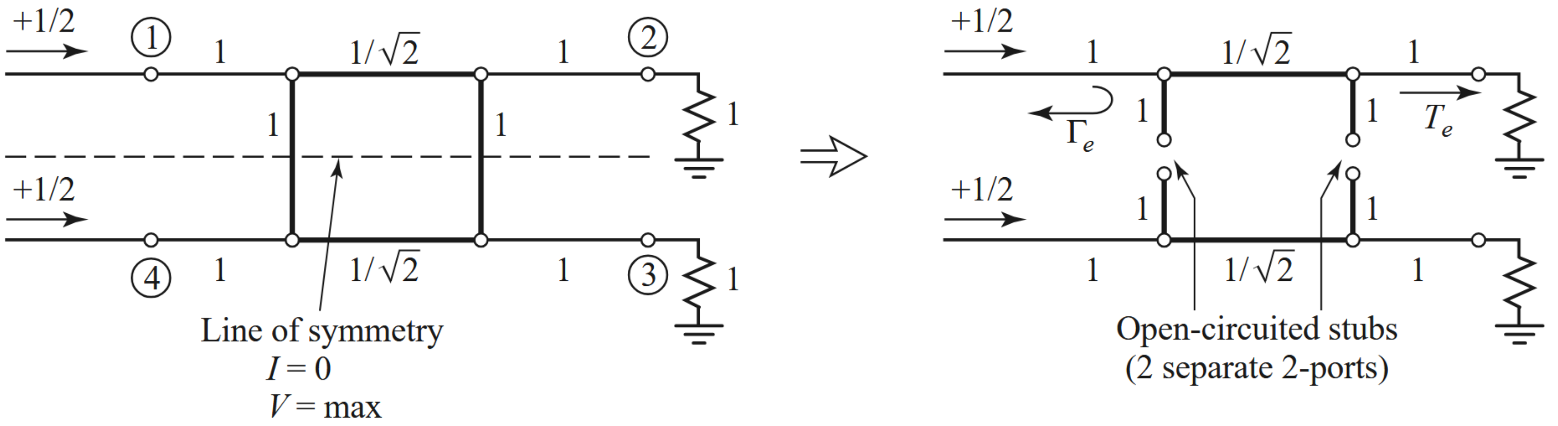




如何推导上述耦合器的S参数？

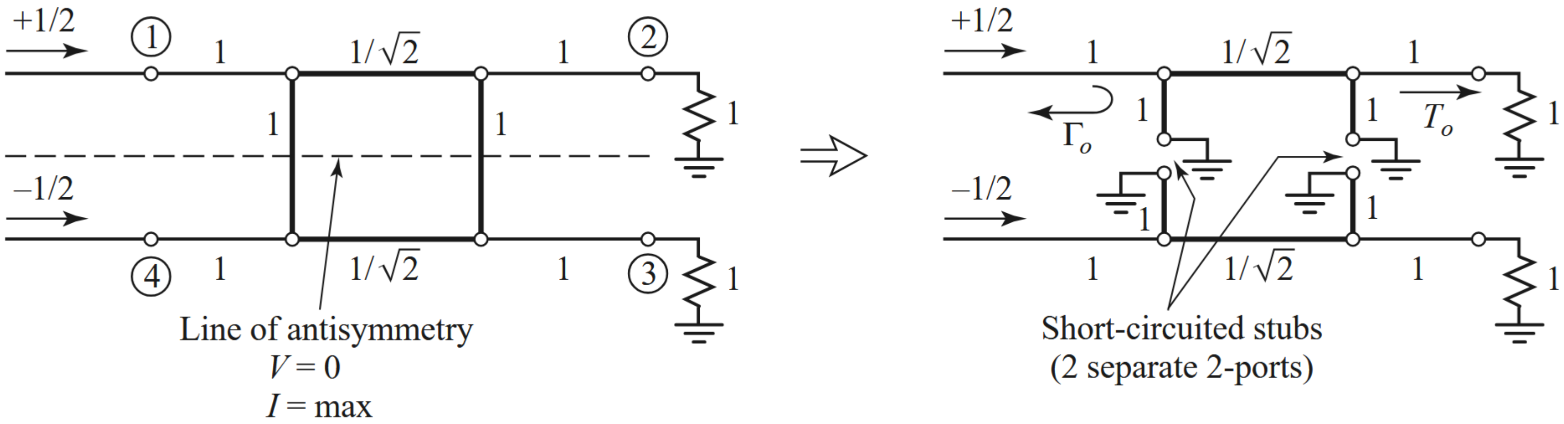
支线耦合器-奇偶模分析

偶模式



(a)

奇模式



(b)

$$B_1 = \frac{1}{2}\Gamma_e + \frac{1}{2}\Gamma_o,$$

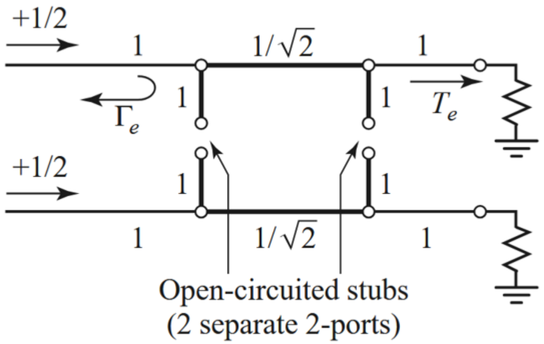
$$B_2 = \frac{1}{2}T_e + \frac{1}{2}T_o,$$

$$B_3 = \frac{1}{2}T_e - \frac{1}{2}T_o,$$

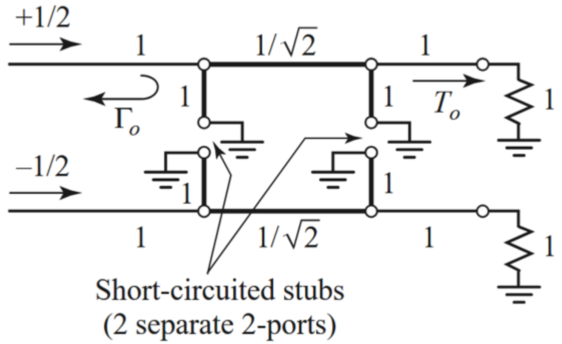
$$B_4 = \frac{1}{2}\Gamma_e - \frac{1}{2}\Gamma_o,$$

支线耦合器-奇偶模分析

偶模式



奇模式



$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}_e = \underbrace{\begin{bmatrix} 1 & 0 \\ j & 1 \end{bmatrix}}_{\substack{\text{Shunt} \\ Y=j}} \underbrace{\begin{bmatrix} 0 & j/\sqrt{2} \\ j\sqrt{2} & 0 \end{bmatrix}}_{\substack{\lambda/4 \\ \text{Transmission} \\ \text{line}}} \underbrace{\begin{bmatrix} 1 & 0 \\ j & 1 \end{bmatrix}}_{\substack{\text{Shunt} \\ Y=J}} = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & j \\ j & -1 \end{bmatrix},$$

$$\Gamma_e = \frac{A+B-C-D}{A+B+C+D} = \frac{(-1+j-j+1)/\sqrt{2}}{(-1+j+j-1)/\sqrt{2}} = 0,$$

$$T_e = \frac{2}{A+B+C+D} = \frac{2}{(-1+j+j-1)/\sqrt{2}} = \frac{-1}{\sqrt{2}}(1+j).$$

同理，

$$\Gamma_o = 0,$$

$$T_o = \frac{1}{\sqrt{2}}(1-j).$$

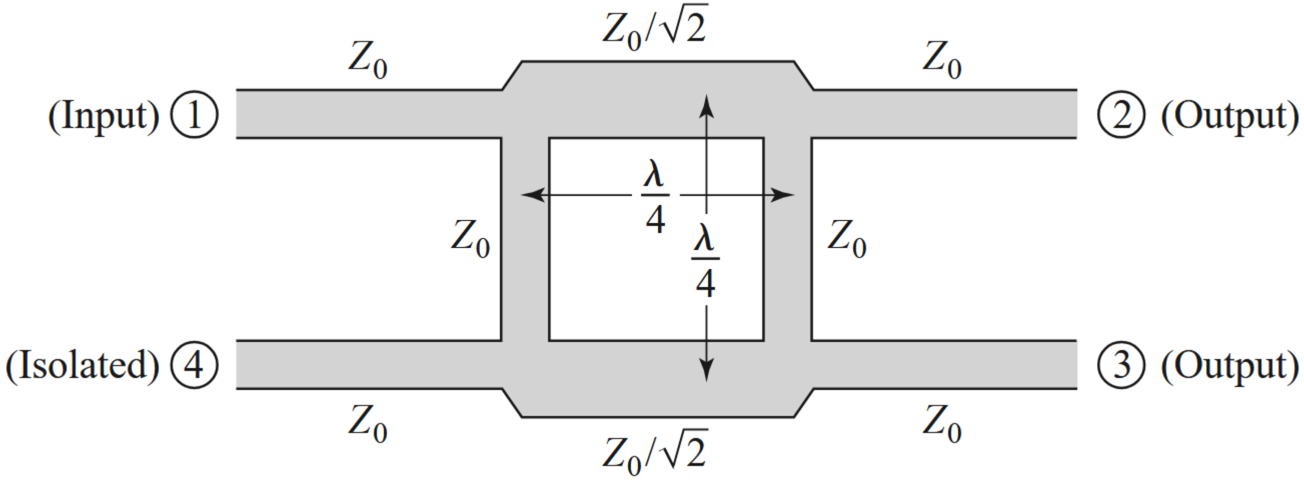
$B_1 = 0$ (port 1 is matched),

$B_2 = -\dfrac{j}{\sqrt{2}}$ (half-power, $-90°$ phase shift from port 1 to 2),

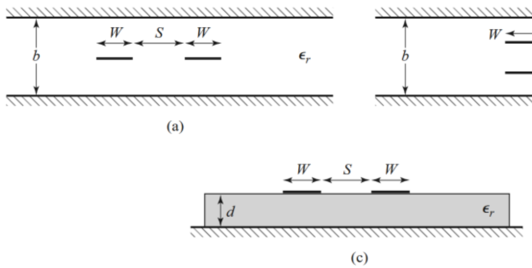
$B_3 = -\dfrac{1}{\sqrt{2}}$ (half-power, $-180°$ phase shift from port 1 to 3),

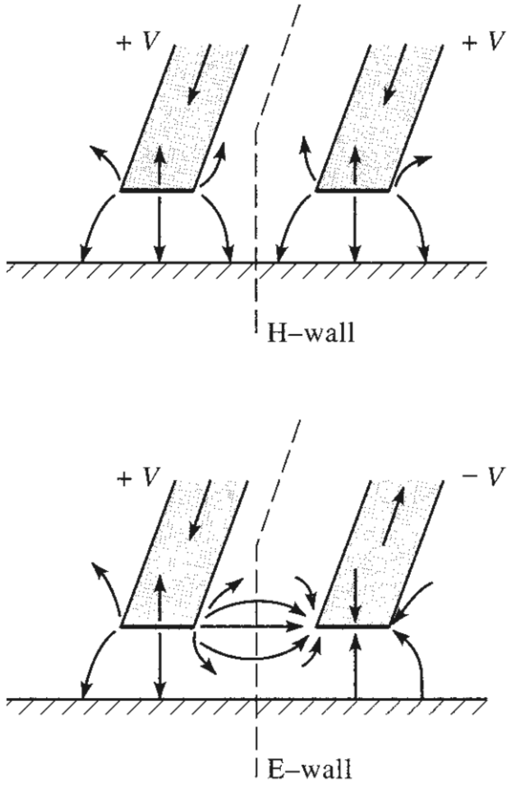
$B_4 = 0$ (no power to port 4).

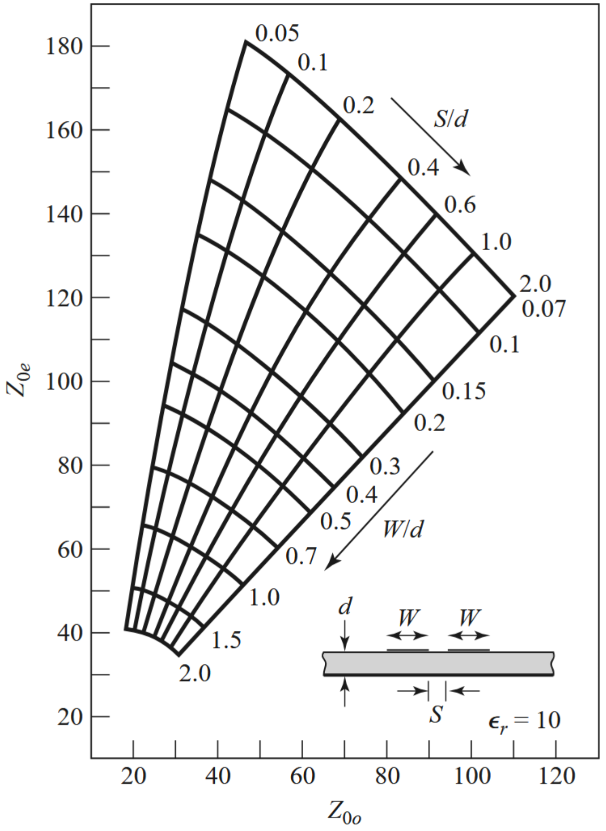
例题：设计一个中心频点为2 GHz的微带线型3-dB正交混合耦合器，所用介质厚度为0.8mm，介电常数为2.3，损耗角为0.0009。分别采用ADS以及HFSS进行设计并对比结果，提供ADS以及HFSS的**器件结构截图**以及仿真的**S参数截图。**



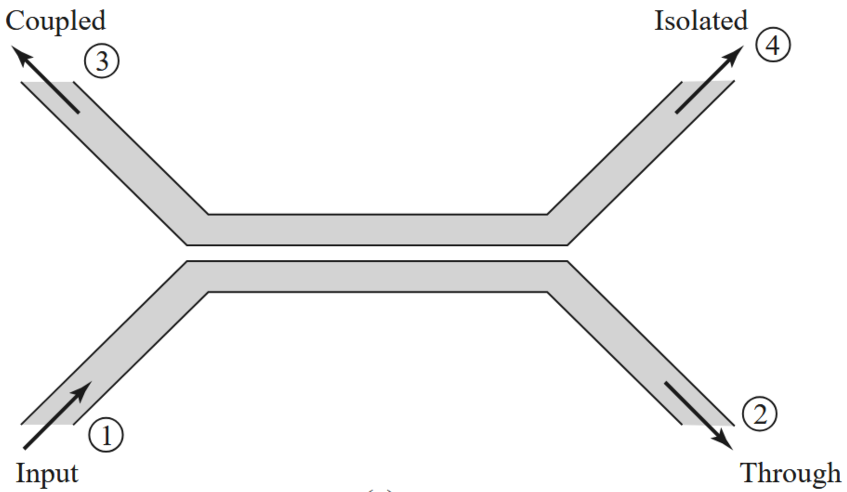
耦合传输线耦合器-耦合传输线



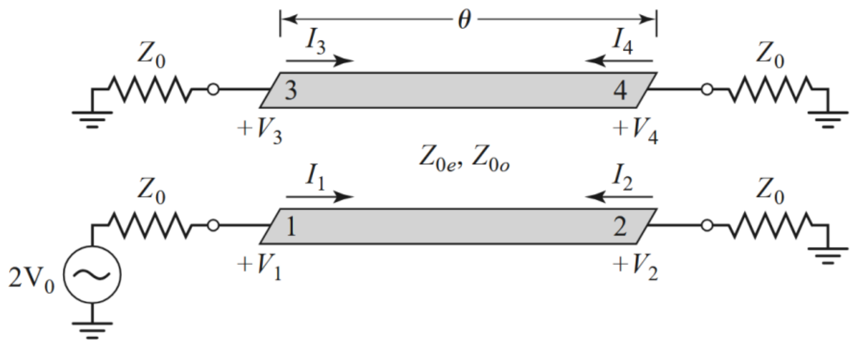




耦合传输线耦合器



(a)



$$V_3 = V_0 \frac{jC\tan\theta}{\sqrt{1-C^2} + j\tan\theta}. \qquad C = \frac{Z_{0e} - Z_{0o}}{Z_{0e} + Z_{0o}},$$

$$V_4 = V_4^e + V_4^o = V_2^e - V_2^o = 0,$$

$$V_2 = V_2^e + V_2^o = V_0 \frac{\sqrt{1-C^2}}{\sqrt{1-C^2}\cos\theta + j\sin\theta}.$$

即长度为四分之一波长

$$\theta = \pi/2$$

$$\frac{V_3}{V_0} = C,$$

$$\frac{V_2}{V_0} = -j\sqrt{1-C^2},$$

If C is specified

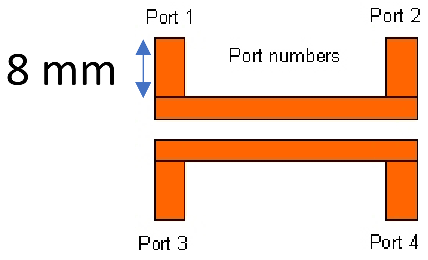
$$Z_{0e} = Z_0\sqrt{\frac{1+C}{1-C}},$$

$$Z_{0o} = Z_0\sqrt{\frac{1-C}{1+C}}.$$

例题：使用**HFSS**设计一个10 dB（C=0.31）单节耦合线耦合器，采用**微带线结构**，地平面间距为1 mm，相对介电常数为2.2，特性阻抗为50 Ω，中心频率为3 GHz。绘制1至5 GHz频率范围内的耦合度与方向性曲线。考虑损耗的影响，假设介质材料的损耗角正切为0.0009，金属设置为PEC，给出设计的结构截图、仿真的S参数以及场分布截图。（提示：由下面公式算出奇模式和偶模式阻抗，再使用ADS软件中的LineCalc中的MCLIN结构计算耦合微带线的结构参数）

$$Z_{0e} = Z_0\sqrt{\frac{1+C}{1-C}},$$

$$Z_{0o} = Z_0\sqrt{\frac{1-C}{1+C}}.$$



参考结构